BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 5 Lab Packet.

Urinary · Reproductive · Nervous Tissue · Brain · Meninges · Spinal Cord · Nerves

PART A	**Lab structure list** Every structure you can be asked to identify on the Module 5 lab exam, chapter by chapter.
PART B	**Charts** Origin, insertion, action, and nerve for every muscle on the list, flow charts for the routes you trace, and reference charts.
PART C	**Lab exam questions** The kinds of questions you could be asked at a station, by chapter.

How to use this packet

In lab

Bring Part A to every lab. Check off a structure only when you can find it on the model, the cadaver, or the slide without looking at a label. Say the name out loud to your team as you point to it.

The charts

For every muscle, the lab exam can ask where it starts (origin), where it ends (insertion), what it does (action), and which nerve supplies it. Cover the chart and quiz yourself one column at a time. Trace each flow chart with your finger on the model, then cover it and say the route from memory.

The questions

Part C shows the kinds of questions that come with a pinned or pointed structure. Practice them out loud with a partner, with the structure in front of you.

PART A

Lab structure list.

These are the lab structures and concepts to learn for the lab portion of the Module 5 exam. Be prepared to identify structures on models, images, the cadaver, and slides.

CHAPTER 1

The Urinary System

KIDNEY MODEL, CADAVER, AND SLIDES

Competencies 133 to 140

The master list runs this system in five steps: the gross kidney, its vasculature, the renal corpuscle, the renal tubule, then the two traces. Follow it in that order and the two traces answer themselves.

☐ Gross anatomy of the urinary system

Say which kidney sits lower and why.

STRUCTURES TO IDENTIFY

- ☐ Right kidney
- ☐ Left kidney
- ☐ Suprarenal (adrenal) gland
- ☐ Renal capsule
- ☐ Renal cortex
- ☐ Renal medulla
- ☐ Renal column
- ☐ Renal pyramid
- ☐ Renal papilla
- ☐ Minor calyx
- ☐ Major calyx
- ☐ Renal pelvis
- ☐ Renal sinus
- ☐ Renal hilum
- ☐ Ureter
- ☐ Urinary bladder
- ☐ Urethra

☐ Vasculature of the kidney

STRUCTURES TO IDENTIFY

- ☐ Renal artery
- ☐ Arcuate artery
- ☐ Cortical radiate artery
- ☐ Afferent arteriole
- ☐ Efferent arteriole
- ☐ Peritubular capillaries
- ☐ Vasa recta
- ☐ Cortical radiate vein
- ☐ Arcuate vein
- ☐ Renal vein

☐ The renal corpuscle

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Glomerulus
- ☐ Glomerular (Bowman's) capsule
- ☐ Parietal layer of the glomerular capsule
- ☐ Visceral layer of the glomerular capsule
- ☐ Podocytes
- ☐ Capsular space

☐ The renal tubule and the juxtaglomerular apparatus

Name the segments in the order the filtrate meets them.

STRUCTURES TO IDENTIFY

- ☐ Proximal convoluted tubule
- ☐ Loop of Henle, descending limb
- ☐ Loop of Henle, ascending limb
- ☐ Distal convoluted tubule
- ☐ Collecting duct
- ☐ Papillary duct
- ☐ Juxtaglomerular apparatus
- ☐ Macula densa cells
- ☐ Granular (juxtaglomerular) cells

☐ **The bladder and urethra**

STRUCTURES TO IDENTIFY

- ☐ Trigone
- ☐ Ureteral openings
- ☐ Internal urethral orifice
- ☐ Detrusor muscle
- ☐ Internal urethral sphincter
- ☐ External urethral sphincter
- ☐ Male urethra
- ☐ Female urethra

☐ **Also identify**

STRUCTURES TO IDENTIFY

- ☐ Loop of Henle (nephron loop)

Trace the path of urine and filtrate from the proximal convoluted tubule to the outside of the body.

Trace the path of blood through the kidney, from the renal artery to the renal vein.

CHAPTER 2

The Male Reproductive System

PELVIS MODEL, CADAVER, AND SLIDES

Competencies 141 to 145

Sort the structures into the gonad, the ducts, the accessory glands, and the supporting structures before you learn them individually. The trace at the end is hers, and it runs straight through the duct group.

☐ **The scrotum and the testis**

STRUCTURES TO IDENTIFY

- ☐ Scrotum
- ☐ Testis
- ☐ Cremaster muscle
- ☐ Dartos muscle
- ☐ Tunica vaginalis
- ☐ Tunica albuginea
- ☐ Seminiferous tubules
- ☐ Testicular artery
- ☐ Testicular vein
- ☐ Pampiniform plexus
- ☐ Spermatic cord
- ☐ Inguinal canal

☐ **Testis tissue, on the slide**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Seminiferous tubule
- ☐ Sustentacular (Sertoli) cells
- ☐ Interstitial (Leydig) cells
- ☐ Developing sperm cells

☐ **The duct system**

Only one structure on this sheet produces sperm. Name it, then say what each of the others does with them.

STRUCTURES TO IDENTIFY

- ☐ Epididymis
- ☐ Vas deferens (ductus deferens)
- ☐ Ejaculatory duct
- ☐ Prostatic urethra
- ☐ Membranous urethra
- ☐ Spongy (penile) urethra

☐ **The accessory glands**

Say what each gland contributes to semen.

STRUCTURES TO IDENTIFY

☐ Seminal vesicles ☐ Prostate gland ☐ Bulbourethral glands

☐ **The penis**

STRUCTURES TO IDENTIFY

☐ Glans penis ☐ Corpus spongiosum ☐ Spongy (penile) urethra

☐ Prepuce ☐ Corpora cavernosa

☐ **The pelvic vessels**

STRUCTURES TO IDENTIFY

☐ Internal iliac artery ☐ Medial umbilical ligament ☐ External iliac vein

☐ External iliac artery ☐ Internal iliac vein

☐ **Also identify**

STRUCTURES TO IDENTIFY

☐ Penis ☐ Urethra

Medial umbilical ligament. It is the remnant of which fetal structure?

Trace the path of sperm from the seminiferous tubules to the outside of the body.

CHAPTER 3

The Female Reproductive System

PELVIS MODEL, CADAVER, AND SLIDES

Competencies 146 to 150

The ligaments are the hard part of this sheet, so name what each one connects as you find it. The tract itself is short and runs in one direction, from the ovary to the vaginal orifice.

☐ **The ovary and its attachments**

Say what each ligament attaches the ovary to.

STRUCTURES TO IDENTIFY

☐ Ovary ☐ Suspensory ligament ☐ Ovarian artery

☐ Ovarian ligament ☐ Broad ligament of the uterus ☐ Ovarian vein

☐ **Ovary tissue, on the slide**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

☐ Ovarian cortex ☐ Ovarian follicles ☐ Corpus luteum

☐ Ovarian medulla ☐ Mature follicle ☐ Corpus albicans

☐ **The uterine tubes**

Name the region where fertilization usually happens.

STRUCTURES TO IDENTIFY

- ☐ Fimbriae
- ☐ Ampulla
- ☐ Infundibulum
- ☐ Isthmus

☐ **The uterus**

Name the layer that sheds in menstruation and the layer that rebuilds it.

STRUCTURES TO IDENTIFY

- ☐ Fundus
- ☐ Body
- ☐ Cervix
- ☐ Uterine cavity
- ☐ Cervical canal
- ☐ Round ligament of the uterus
- ☐ Broad ligament of the uterus
- ☐ Perimetrium
- ☐ Myometrium
- ☐ Endometrium

☐ **The vagina and the vulva**

STRUCTURES TO IDENTIFY

- ☐ Vagina
- ☐ Vaginal orifice
- ☐ Fornix
- ☐ Mons pubis
- ☐ Labia majora
- ☐ Labia minora
- ☐ Clitoris
- ☐ Urethral orifice
- ☐ Vestibule
- ☐ Perineum

☐ **The pelvic vessels**

STRUCTURES TO IDENTIFY

- ☐ Internal iliac artery
- ☐ External iliac artery
- ☐ Medial umbilical ligament
- ☐ Internal iliac vein
- ☐ External iliac vein

Medial umbilical ligament. It is the remnant of which fetal structure?

Trace the path of an oocyte from the ovary to the uterine cavity.

CHAPTER 4

Nervous Tissue, Histology

THE SLIDES AND THE NEURON MODEL

Competencies 173 to 178

Five slides, and the master list names all five. Fill in function and location as you go, because on this sheet the cell you are pointing at is often identified by the tissue it sits in rather than by its own shape.

☐ **Multipolar neuron, on the slide and the model**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Cell body (soma)
- ☐ Nucleus
- ☐ Nucleolus
- ☐ Cell processes (dendrites)
- ☐ Axon
- ☐ Axon hillock
- ☐ Synaptic bulb (terminal end)
- ☐ Nissl bodies
- ☐ Neurofibrils

☐ **Spinal cord, cross section**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ White matter
- ☐ Gray matter
- ☐ Ventral (anterior) horn
- ☐ Dorsal (posterior) horn
- ☐ Lateral horn, if present
- ☐ Central canal

☐ **Cerebral cortex**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Pyramidal cells
- ☐ Gray matter of the cortex
- ☐ Underlying white matter

☐ **Cerebellum**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Purkinje cells
- ☐ Cerebellar cortex
- ☐ Arbor vitae

☐ **Nerve, cross section**

All of these are best seen on a cross section. Work outward from a single axon to the whole nerve.

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

- ☐ Axon
- ☐ Schwann cells
- ☐ Nodes of Ranvier
- ☐ Endoneurium
- ☐ Perineurium
- ☐ Epineurium
- ☐ Fascicle

☐ **The neuroglia**

Sort the six into the four of the central nervous system and the two of the peripheral nervous system, and give the job of each.

STRUCTURES TO IDENTIFY

- ☐ Astrocytes
- ☐ Oligodendrocytes
- ☐ Microglia
- ☐ Ependymal cells
- ☐ Schwann cells
- ☐ Satellite cells

A cluster of cell bodies is called a _____ in the CNS and a _____ in the PNS.

A bundle of axons is called a _____ in the CNS and a _____ in the PNS.

Which cell builds the myelin in the PNS, and which one builds it in the CNS?

_____

CHAPTER 5

The Brain

THE BRAIN MODEL, THE SHEEP BRAIN, AND THE CADAVER

Competencies 151 to 160

The largest identification block of the module. Work it surface first, then midsagittal, then inferior, because that is the order the models are set out in and it is how the tags follow.

☐ Surface features of the cerebrum

Know the functional regions of pre & post central gyrus.

STRUCTURES TO IDENTIFY

- ☐ Gyrus
- ☐ Sulcus
- ☐ Longitudinal fissure
- ☐ Transverse fissure
- ☐ Precentral gyrus
- ☐ Precentral sulcus
- ☐ Central sulcus
- ☐ Postcentral gyrus
- ☐ Postcentral sulcus
- ☐ Lateral sulcus (Sylvian fissure)

☐ The lobes

Name the sulcus or fissure that bounds each lobe.

STRUCTURES TO IDENTIFY

- ☐ Frontal lobe
- ☐ Parietal lobe
- ☐ Temporal lobe
- ☐ Occipital lobe
- ☐ Insula

☐ Midsagittal section

STRUCTURES TO IDENTIFY

- ☐ Corpus callosum
- ☐ Sulcus of the corpus callosum
- ☐ Cingulate gyrus
- ☐ Cingulate sulcus
- ☐ Septum pellucidum
- ☐ Fornix
- ☐ Thalamus
- ☐ Massa intermedia (interthalamic adhesion)
- ☐ Hypothalamus
- ☐ Infundibulum
- ☐ Pineal gland
- ☐ Optic chiasm
- ☐ Cerebellum
- ☐ Transverse fissure

☐ Inferior surface

STRUCTURES TO IDENTIFY

- ☐ Uncus
- ☐ Parahippocampal gyrus
- ☐ Mammillary bodies
- ☐ Crus cerebri
- ☐ Optic chiasm
- ☐ Optic nerve
- ☐ Optic tract
- ☐ Olfactory bulb
- ☐ Olfactory tract

☐ Functional areas of the cortex

Name the gyrus each cortical area sits on, and say which hemisphere Broca's and Wernicke's areas are usually found in.

STRUCTURES TO IDENTIFY

- ☐ Primary motor cortex
- ☐ Primary sensory cortex
- ☐ Broca's area
- ☐ Wernicke's area
- ☐ Thalamus

☐ The cerebellum

STRUCTURES TO IDENTIFY

- ☐ Cerebellar hemispheres
- ☐ Vermis
- ☐ Folia
- ☐ Cerebellar cortex
- ☐ Arbor vitae
- ☐ Transverse fissure

☐ **The cerebral arterial circle (Circle of Willis)**

Trace the ring from a vertebral artery around to an anterior cerebral artery, naming every vessel on the way.

STRUCTURES TO IDENTIFY

- ☐ Internal carotid artery
- ☐ Anterior cerebral artery
- ☐ Anterior communicating artery
- ☐ Middle cerebral artery
- ☐ Posterior communicating artery
- ☐ Posterior cerebral artery
- ☐ Basilar artery
- ☐ Pontine arteries
- ☐ Vertebral arteries
- ☐ Superior cerebellar artery
- ☐ Anterior inferior cerebellar artery
- ☐ Posterior inferior cerebellar artery
- ☐ Anterior spinal artery

Which gyrus lies immediately anterior to the central sulcus, and what does it do?

CHAPTER 6

The Brainstem and Cerebellum

THE BRAIN MODEL AND THE SHEEP BRAIN

Competencies 161 to 166

Three regions from the top down, plus the cerebellum hanging off the back of them. Every structure here is either a surface landmark or a stalk, so name the region before you name the part.

☐ **The three regions**

Name the cranial nerves that arise from each region.

STRUCTURES TO IDENTIFY

- ☐ Midbrain
- ☐ Pons
- ☐ Medulla oblongata
- ☐ Pontomedullary junction

☐ **The midbrain**

STRUCTURES TO IDENTIFY

- ☐ Cerebral peduncles
- ☐ Crus cerebri
- ☐ Cerebral aqueduct

☐ **The pons**

STRUCTURES TO IDENTIFY

- ☐ Basilar pons
- ☐ Fourth ventricle

☐ **The medulla oblongata**

Say what crosses at the decussation of the pyramids and what that crossing explains.

STRUCTURES TO IDENTIFY

- ☐ Pyramids
- ☐ Decussation of the pyramids
- ☐ Medullary olive
- ☐ Pontomedullary junction

☐ **The cerebellum and its peduncles**

Say which region of the brainstem each peduncle connects to.

STRUCTURES TO IDENTIFY

- ☐ Cerebellar hemispheres
- ☐ Vermis
- ☐ Folia
- ☐ Arbor vitae
- ☐ Superior cerebellar peduncle
- ☐ Middle cerebellar peduncle
- ☐ Inferior cerebellar peduncle

☐ **Also identify**

STRUCTURES TO IDENTIFY

- ☐ Corpora quadrigemina
- ☐ Superior colliculi
- ☐ Inferior colliculi
- ☐ Cerebellum

Which region of the brainstem is continuous with the spinal cord, and at what landmark?

CHAPTER 7

Meninges, Ventricles, and CSF

THE BRAIN MODEL AND THE CORONAL SECTION

Competencies 167 to 172

Three membranes, three spaces, four ventricles, and one circuit. The master list asks you to trace the fluid from the lateral ventricles to reabsorption, and it warns that some structures on that route are named but not seen on the model.

☐ **The cranial meninges**

Name the three layers in order from the skull inward.

STRUCTURES TO IDENTIFY

- ☐ Dura mater
- ☐ Periosteal layer of the dura
- ☐ Meningeal layer of the dura
- ☐ Arachnoid mater
- ☐ Pia mater

☐ **The dural extensions**

For each one, say where it is found and which structures it separates.

STRUCTURES TO IDENTIFY

- ☐ Falx cerebri
- ☐ Falx cerebelli
- ☐ Tentorium cerebelli

☐ **The dural venous sinuses**

Say where the sigmoid sinus leaves the skull and what the blood becomes once it does.

STRUCTURES TO IDENTIFY

- ☐ Superior sagittal sinus
- ☐ Inferior sagittal sinus
- ☐ Straight sinus
- ☐ Confluence of sinuses
- ☐ Transverse sinus
- ☐ Sigmoid sinus

☐ **The meningeal spaces**

Say which of these spaces holds cerebrospinal fluid and which is only a potential space.

STRUCTURES TO IDENTIFY

- ☐ Epidural space
- ☐ Subdural space
- ☐ Subarachnoid space
- ☐ Arachnoid granulations

☐ **The ventricles**

STRUCTURES TO IDENTIFY

- ☐ Lateral ventricle
- ☐ Interventricular foramen of Monro
- ☐ Third ventricle
- ☐ Cerebral aqueduct
- ☐ Fourth ventricle
- ☐ Central canal
- ☐ Septum pellucidum

☐ **The choroid plexus, on the slide and the model**

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

☐ Choroid plexus ☐ Ependymal cells

Trace cerebrospinal fluid from the lateral ventricles to its reabsorption, naming every structure it passes. Include the median and lateral apertures and the central canal even though they are not seen on the model.

_____

_____

CHAPTER 8

The Spinal Cord

THE CORD MODEL, THE CROSS SECTION, AND THE CADAVER

Competencies 184 to 186

The cord in two views. The whole cord from the outside gives you the enlargements and the terminations; the cross section gives you the horns, the commissure, and the canal.

☐ **The spinal meninges**

Say what fills the epidural space in the vertebral canal, and why there is no epidural space inside the cranium.

STRUCTURES TO IDENTIFY

☐ Dura mater ☐ Epidural space ☐ Denticulate ligaments
☐ Arachnoid mater ☐ Subdural space
☐ Pia mater ☐ Subarachnoid space

☐ **External anatomy of the cord**

Give the vertebral level where the cord ends, and say what fills the canal below that point.

STRUCTURES TO IDENTIFY

☐ Cervical enlargement ☐ Cauda equina ☐ Posterior median sulcus
☐ Lumbar enlargement ☐ Filum terminale
☐ Conus medullaris ☐ Anterior median fissure

☐ **The cord in cross section**

Say which segments of the cord carry a lateral horn and what sits in it.

Function _____ **Location** _____

Special structures _____

STRUCTURES TO IDENTIFY

☐ Ventral (anterior) horn ☐ White matter ☐ Anterior column
☐ Lateral horn ☐ Gray commissure ☐ Lateral column
☐ Dorsal (posterior) horn ☐ Central canal ☐ Posterior column
☐ Gray matter ☐ Anterior white commissure

Where in the vertebral column is a lumbar puncture performed, and why is that level safe?

_____

CHAPTER 9

Peripheral Nerves and Plexuses

THE CORD MODEL, THE CADAVER, AND THE TORSO

Competencies 187 to 189, 191

The spinal nerve is the hinge of this sheet. Everything before it is a root, everything after it is a ramus, and the list asks you to say what kind of information travels in each.

☐ **The spinal nerve and its roots**

For each structure say what it carries: sensory axons, motor axons, mixed, or sensory cell bodies.

STRUCTURES TO IDENTIFY

- ☐ Ventral (anterior) root
- ☐ Dorsal (posterior) root
- ☐ Dorsal root ganglion
- ☐ Spinal nerve
- ☐ Intervertebral foramen
- ☐ Rootlets

☐ **The rami and the sympathetic connections**

Say which ramus forms the plexuses and which one serves the deep muscles and skin of the back.

STRUCTURES TO IDENTIFY

- ☐ Ventral (anterior) ramus
- ☐ Dorsal (posterior) ramus
- ☐ Gray ramus communicans
- ☐ White ramus communicans
- ☐ Sympathetic chain ganglion
- ☐ Sympathetic trunk

☐ **The plexuses**

Give the spinal nerves that form each plexus and the region of the body each one serves.

STRUCTURES TO IDENTIFY

- ☐ Cervical plexus
- ☐ Brachial plexus
- ☐ Lumbar plexus
- ☐ Sacral plexus

☐ **Nerves to find on the cadaver and the models**

Name the plexus the phrenic nerve comes from and the structure it supplies.

STRUCTURES TO IDENTIFY

- ☐ Phrenic nerve

Which root carries a ganglion, and whose cell bodies are in it?

A patient loses sensation and strength across one whole upper limb. Which plexus is damaged?

CHAPTER 10

The Cranial Nerves

THE BRAIN MODEL, THE SKULL, AND THE SHEEP BRAIN

Competency 190

Twelve nerves, and the master list asks four things of each one: identify it, classify it as sensory, motor, or both, give its function, and find the opening it leaves the skull through. Do all four every time you point at one.

☐ The twelve cranial nerves, I to XII

For each nerve give the number, the name, the classification as sensory, motor, or both, and the function.

STRUCTURES TO IDENTIFY

- ☐ CN I, Olfactory
- ☐ CN II, Optic
- ☐ CN III, Oculomotor
- ☐ CN IV, Trochlear
- ☐ CN V, Trigeminal
- ☐ CN VI, Abducens
- ☐ CN VII, Facial
- ☐ CN VIII, Vestibulocochlear
- ☐ CN IX, Glossopharyngeal
- ☐ CN X, Vagus
- ☐ CN XI, Accessory
- ☐ CN XII, Hypoglossal

☐ Structures associated with the cranial nerves

Name the nerve that passes through each opening.

STRUCTURES TO IDENTIFY

- ☐ Pontomedullary junction
- ☐ Internal acoustic meatus
- ☐ Olfactory bulb
- ☐ Olfactory tract
- ☐ Olfactory foramina in the cribriform plate
- ☐ Optic nerve
- ☐ Optic chiasm
- ☐ Optic tract
- ☐ Superior orbital fissure
- ☐ Optic canal

☐ The extrinsic eye muscles

Name the cranial nerve that supplies each muscle. Three nerves cover all six.

STRUCTURES TO IDENTIFY

- ☐ Lateral rectus
- ☐ Medial rectus
- ☐ Superior rectus
- ☐ Inferior rectus
- ☐ Superior oblique
- ☐ Inferior oblique

Which three cranial nerves are purely sensory?

_____

Which four cranial nerves are mixed?

_____

PART B

Charts.

Muscle charts first, grouped the way you will find the muscles in lab, then the flow charts and reference charts.

SCAN TO DRILL THESE CHARTS

Interactive Muscle Charts: I, O, A

Quiz yourself on the origin, insertion, action, and nerve for each muscle on your phone.

In the app, this module's muscles are under Eye (Extrinsic).

medmasterscollaborative.com/muscle-charts-i-o-a-inn

The extrinsic eye muscles

LR6 SO4: the lateral rectus is supplied by CN VI and the superior oblique by CN IV. The other four are supplied by CN III.

MUSCLE	ORIGIN	INSERTION	ACTION	NERVE
Lateral rectus	Common tendinous ring at the back of the orbit	Lateral surface of the eyeball	Moves the eye laterally (abducts)	Abducens nerve (CN VI)
Medial rectus	Common tendinous ring	Medial surface of the eyeball	Moves the eye medially (adducts)	Oculomotor nerve (CN III)
Superior rectus	Common tendinous ring	Superior surface of the eyeball	Moves the eye up and toward the midline	Oculomotor nerve (CN III)
Inferior rectus	Common tendinous ring	Inferior surface of the eyeball	Moves the eye down and toward the midline	Oculomotor nerve (CN III)
Superior oblique	Body of the sphenoid, above and medial to the optic canal; its tendon passes through the trochlea, a pulley on the frontal bone	Superior lateral surface of the eyeball, behind its middle	Moves the eye down and laterally	Trochlear nerve (CN IV)
Inferior oblique	Floor of the orbit at the front, on the maxilla	Inferior lateral surface of the eyeball, behind its middle	Moves the eye up and laterally	Oculomotor nerve (CN III)

Blood flow through the kidney

Blood passes through two capillary beds in a row: the glomerulus, where it is filtered, and then the peritubular capillaries and vasa recta around the tubules. The efferent arteriole is the vessel between them.

1. **Renal artery** Branches from the abdominal aorta and enters the kidney at the renal hilum
2. **Segmental arteries** Branches of the renal artery in the renal sinus

3. **Interlobar arteries** Run outward through the renal columns, between the renal pyramids
4. **Arcuate arteries** Arch over the bases of the renal pyramids, at the border between the cortex and the medulla
5. **Cortical radiate arteries** Also called interlobular arteries. They run straight out into the renal cortex
6. **Afferent arteriole** Carries blood into the glomerulus
7. **Glomerulus** The first capillary bed. Blood is filtered here into the glomerular capsule
8. **Efferent arteriole** Carries the filtered blood out of the glomerulus
9. **Peritubular capillaries and vasa recta** The second capillary bed. Peritubular capillaries wrap the tubules in the cortex; the vasa recta run beside the long loops of Henle in the medulla
10. **Cortical radiate veins** Collect blood from the cortex and run back toward the medulla
11. **Arcuate veins** Run along the bases of the renal pyramids beside the arcuate arteries
12. **Interlobar veins** Run back through the renal columns toward the renal sinus
13. **Renal vein** Leaves the kidney at the renal hilum and drains into the inferior vena cava

The path of filtrate and urine

The fluid is called filtrate while it is inside the nephron and urine once it leaves the papillary duct. The nephron and collecting duct change the fluid; everything after the renal papilla only carries and stores urine.

1. **Glomerular capsule (capsular space)** Filtrate leaves the glomerulus and collects here
2. **Proximal convoluted tubule** The first coiled segment, in the renal cortex
3. **Loop of Henle, descending limb** Dips down into the renal medulla
4. **Loop of Henle, ascending limb** Climbs back up toward the cortex
5. **Distal convoluted tubule** The second coiled segment, in the cortex. It touches its own glomerulus at the juxtaglomerular apparatus
6. **Collecting duct** Gathers fluid from many nephrons and runs down through a renal pyramid
7. **Papillary duct** The last part of the collecting duct. It opens at the tip of the renal papilla
8. **Minor calyx** Catches urine from one renal papilla
9. **Major calyx** Formed where several minor calyces join
10. **Renal pelvis** Funnels urine to the ureter at the hilum
11. **Ureter** Carries urine down to the bladder by peristalsis
12. **Urinary bladder** Urine enters through the ureteral openings in the trigone and is stored
13. **Urethra** Urine leaves through the internal urethral orifice, passes the internal and external urethral sphincters, and exits at the external urethral orifice

The path of sperm

Only the seminiferous tubules make sperm. Every other structure stores them, matures them, or carries them. The seminal vesicles add their fluid at the ejaculatory duct, the prostate in the prostatic urethra, and the bulbourethral glands in the spongy urethra.

1. **Seminiferous tubules** Sperm are made here, inside the testis. Straight tubules carry them into the rete testis, and the efferent ductules carry them out of the testis into the epididymis

2. **Epididymis** The coiled duct on the back of the testis where sperm are stored and gain the ability to swim
3. **Vas deferens (ductus deferens)** Runs up in the spermatic cord, through the inguinal canal, and over the back of the urinary bladder
4. **Ejaculatory duct** Formed where the vas deferens joins the duct of the seminal vesicle. It passes through the prostate
5. **Prostatic urethra** The part of the urethra inside the prostate gland
6. **Membranous urethra** The short part that crosses the pelvic floor, through the external urethral sphincter
7. **Spongy (penile) urethra** Runs through the corpus spongiosum to the external urethral orifice at the tip of the glans penis

The path of the ovum

The cell released at ovulation is a secondary oocyte, usually called the ovum or egg. It cannot move on its own; the fimbriae, the cilia of the tube lining, and the smooth muscle of the tube move it along.

1. **Ovary** A mature follicle in the ovarian cortex ruptures at ovulation and releases the oocyte
2. **Pelvic (peritoneal) cavity** The oocyte is released into the pelvic cavity, because the uterine tube is not sealed onto the ovary
3. **Fimbriae** The finger-like fringe at the end of the tube sweeps the oocyte in
4. **Infundibulum** The funnel-shaped open end of the uterine tube
5. **Ampulla** The widest and longest part of the tube, where fertilization usually happens
6. **Isthmus** The narrow part of the tube that joins the uterus
7. **Uterine cavity** The space inside the body of the uterus. If the oocyte was fertilized, the embryo implants in the endometrium

The cerebral arterial circle, from a vertebral artery to an anterior cerebral artery

The ring joins the blood from the internal carotids with the blood from the vertebral arteries, so blood has a second route if one vessel slowly narrows.

1. **Vertebral artery** Enters the skull through the foramen magnum and runs up the front of the medulla
2. **Basilar artery** Formed where the two vertebral arteries join, at the lower edge of the pons. It runs up the front of the pons
3. **Posterior cerebral artery** The basilar artery ends by splitting into the right and left posterior cerebral arteries
4. **Posterior communicating artery** Joins the posterior cerebral artery to the internal carotid artery
5. **Internal carotid artery** Brings blood up from the neck; it gives off the middle cerebral artery
6. **Anterior cerebral artery** Continues forward from the internal carotid artery into the longitudinal fissure
7. **Anterior communicating artery** Joins the right and left anterior cerebral arteries and closes the ring

Venous blood through the dural sinuses

The dural sinuses form where the two layers of the dura separate. They also receive the CSF returned through the arachnoid granulations.

1. **Superior sagittal sinus** Runs along the top edge of the falx cerebri; receives CSF from the arachnoid granulations. It drains straight back to the confluence of sinuses, not into the inferior sagittal sinus
2. **Inferior sagittal sinus** A second, smaller starting point. It runs in the lower free edge of the falx cerebri and drains into the straight sinus

3. **Straight sinus** Runs where the falx cerebri meets the tentorium cerebelli
4. **Confluence of sinuses** Where the superior sagittal sinus and the straight sinus meet, at the back of the skull (internal occipital protuberance)
5. **Transverse sinuses** Run to the right and left from the confluence along the edge of the tentorium cerebelli
6. **Sigmoid sinuses** S-shaped sinuses that curve down to the jugular foramen
7. **Internal jugular vein** The blood leaves the skull through the jugular foramen as the internal jugular vein

CSF circulation

The median aperture, the lateral apertures, and the central canal are on this route but are not seen on the model. CSF is made inside the ventricles and reabsorbed outside them.

1. **Choroid plexus of the lateral ventricles** Ependymal cells over the choroid plexus make CSF in each lateral ventricle
2. **Interventricular foramen of Monro** CSF passes from each lateral ventricle into the third ventricle
3. **Third ventricle** In the diencephalon; its choroid plexus adds more CSF
4. **Cerebral aqueduct** CSF flows down this channel through the midbrain
5. **Fourth ventricle** Behind the pons; its choroid plexus adds more CSF
6. **Central canal (side branch)** The central canal is continuous with the fourth ventricle, so a small amount of CSF flows from the fourth ventricle down into the central canal of the spinal cord
7. **Median aperture and two lateral apertures** Most CSF leaves the fourth ventricle through these three openings
8. **Subarachnoid space** CSF bathes the whole surface of the brain and spinal cord
9. **Arachnoid granulations** CSF passes through the arachnoid granulations into the superior sagittal sinus
10. **Superior sagittal sinus** CSF joins the venous blood and drains with it out of the skull

The path of smell

1. **Olfactory receptor cells** In the lining of the roof of the nasal cavity
2. **Olfactory nerve (CN I)** Its fibers pass up through the olfactory foramina in the cribriform plate
3. **Olfactory bulb** The fibers end here, on top of the cribriform plate
4. **Olfactory tract** Carries the signal back under the frontal lobe
5. **Olfactory cortex** In the medial temporal lobe, at the uncus

The path of vision

Because fibers from the medial half of each retina cross at the optic chiasm, each optic tract carries the view of the opposite side of the world.

1. **Retina** Photoreceptors in the back of the eye detect light
2. **Optic nerve (CN II)** Leaves the back of the eye and passes through the optic canal
3. **Optic chiasm** Fibers from the medial half of each retina cross to the other side
4. **Optic tract** Carries the fibers from the chiasm back to the thalamus

5. **Thalamus** Relays the signal on to the cortex
6. **Occipital lobe** The primary visual cortex, where you see the image

Gross anatomy of the kidney

Right and left kidney
Retroperitoneal, between T12 and L3. The right kidney sits a little lower because the liver is above it.

Suprarenal (adrenal) gland
The endocrine gland that sits on the superior pole of each kidney.

Renal capsule
The smooth fibrous layer on the surface of the kidney that keeps its shape.

Renal cortex
The outer region of kidney tissue, just under the capsule. It holds the renal corpuscles and the convoluted tubules.

Renal medulla
The inner region of kidney tissue, made up of the renal pyramids.

Renal pyramid
A cone-shaped mass of the medulla, with its base toward the cortex and its tip pointing toward the renal sinus.

Renal column
An extension of the cortex that dips down between two renal pyramids.

Renal papilla
The tip of a renal pyramid. Urine drips from it into a minor calyx.

Minor calyx
The small cup that catches urine from one renal papilla.

Major calyx
The larger cup formed where several minor calyces join.

Renal pelvis
The funnel formed by the major calyces. It narrows into the ureter at the hilum.

Renal sinus
The space inside the kidney, just in from the hilum, that holds the renal pelvis, the calyces, the vessels, and fat.

Renal hilum
The notch on the medial border where the renal artery, renal vein, ureter, and nerves enter and leave.

Ureter
The muscular tube that carries urine from the renal pelvis to the bladder by peristalsis.

Urinary bladder
The muscular sac in the pelvis that stores urine.

Urethra
The tube that carries urine from the bladder to the outside.

The renal corpuscle and the juxtaglomerular apparatus

Renal corpuscle
The filtering part of the nephron: the glomerulus inside the glomerular capsule. Found in the renal cortex.

Glomerulus
The tuft of fenestrated capillaries where blood is filtered.

Glomerular (Bowman's) capsule
The double-walled cup around the glomerulus that collects the filtrate.

Parietal layer of the glomerular capsule
The outer wall of the capsule, made of simple squamous epithelium.

Visceral layer of the glomerular capsule
The inner layer that clings to the glomerular capillaries. It is made of podocytes.

Podocytes
The cells of the visceral layer. Their foot processes leave filtration slits between them.

Capsular space
The space between the parietal and visceral layers, where filtrate collects before it enters the proximal convoluted tubule.

Macula densa cells
Crowded cells in the wall of the distal convoluted tubule that sense the sodium and chloride in the fluid.

Juxtaglomerular apparatus
The spot where the distal convoluted tubule touches the afferent and efferent arterioles of its own glomerulus. It helps control filtration and blood pressure.

Granular (juxtaglomerular) cells
Modified smooth muscle cells in the wall of the afferent arteriole that release renin when blood pressure falls.

The bladder and the urethra

STRUCTURE	WHAT IT IS	WHAT TO KNOW
Trigone	The smooth triangle on the floor of the bladder	Its three corners are the two ureteral openings and the internal urethral orifice
Ureteral openings	Where the two ureters open into the bladder	The two upper corners of the trigone
Internal urethral orifice	Where the urethra leaves the bladder	The lower corner of the trigone
Detrusor muscle	The smooth muscle of the bladder wall	Contracts to empty the bladder
Internal urethral sphincter	A ring of smooth muscle at the bladder outlet	Involuntary
External urethral sphincter	A ring of skeletal muscle in the pelvic floor around the urethra	Voluntary; gives conscious control of urination
Male urethra	About 20 cm long	Three parts: prostatic, membranous, and spongy. It carries both urine and semen
Female urethra	About 4 cm long	Opens in the vestibule between the clitoris and the vaginal orifice. Its short length is one reason urinary tract infections are more common in women

The scrotum and the testis

Scrotum
The pouch of skin that holds the testes outside the body, where they stay a few degrees cooler than the core.

Cremaster muscle
Skeletal muscle that runs down the spermatic cord. It contracts to raise the testes toward the body when it is cold and relaxes to let them drop when it is warm.

Tunica albuginea
The white fibrous capsule of the testis. It sends partitions inward that divide the testis into lobules.

Testicular artery
Branches from the abdominal aorta and reaches the testis through the spermatic cord.

Pampiniform plexus
The network of veins around the testicular artery in the spermatic cord. It cools the arterial blood before it reaches the testis.

Inguinal canal
The passage through the lower anterior abdominal wall that carries the spermatic cord.

Dartos muscle
Smooth muscle in the wall of the scrotum. It wrinkles the scrotal skin to reduce heat loss.

Tunica vaginalis
The serous membrane, from the peritoneum, that partly covers the front and sides of each testis.

Seminiferous tubules
The tightly coiled tubules inside the lobules where sperm are produced.

Testicular vein
Drains the testis. Inside the spermatic cord it forms the pampiniform plexus.

Spermatic cord
The bundle that holds the vas deferens, the testicular artery, the pampiniform plexus, nerves, lymphatic vessels, and the cremaster muscle.

Testis tissue, on the slide

Find the round cut tubules first. Sperm cells develop from the outer edge of each tubule toward its lumen.

STRUCTURE	WHERE IT IS ON THE SLIDE	JOB
Seminiferous tubule	The round or oval tubes cut in cross section	Where sperm are made
Sustentacular (Sertoli) cells	Tall cells that reach from the outer edge of the tubule to the lumen, between the developing sperm cells	Nourish developing sperm and form the blood-testis barrier
Interstitial (Leydig) cells	Small clusters in the connective tissue between the tubules	Make testosterone
Developing sperm cells	Layers in the tubule wall: the least mature at the outer edge, the most mature (with tails) near the lumen	Become sperm

The accessory glands

Semen is sperm plus the fluid from these three sets of glands.

GLAND	WHERE IT IS	WHAT IT ADDS TO SEMEN
Seminal vesicles	Paired glands on the back of the urinary bladder	Alkaline fluid rich in fructose that feeds the sperm; the largest share of semen
Prostate gland	A single gland below the bladder that surrounds the prostatic urethra	Milky fluid that helps the sperm stay active
Bulbourethral glands	Pea-sized paired glands beside the membranous urethra; their ducts open into the spongy urethra	Clear alkaline mucus that neutralizes leftover urine in the urethra and lubricates

The penis

Penis
The organ that carries the spongy urethra and delivers semen.

Prepuce
The foreskin, the loose fold of skin over the glans.

Corpus spongiosum
The single column of erectile tissue on the ventral side. It surrounds the spongy urethra and forms the glans.

Glans penis
The enlarged tip of the penis, formed by the end of the corpus spongiosum.

Corpora cavernosa
The two columns of erectile tissue on the dorsal side of the penis.

The pelvic vessels

The same vessels are on the male and the female list.

VESSEL	WHAT TO KNOW
Internal iliac artery	Branch of the common iliac artery that supplies the pelvic organs, the buttock, and the perineum
External iliac artery	Branch of the common iliac artery that runs along the pelvic brim and becomes the femoral artery as it passes under the inguinal ligament
Medial umbilical ligament	A fibrous cord on the inside of the anterior abdominal wall. It is the remnant of the fetal umbilical artery
Internal iliac vein	Drains the pelvic organs and joins the external iliac vein to form the common iliac vein
External iliac vein	Continues from the femoral vein of the lower limb and joins the internal iliac vein

The ovary and its ligaments

STRUCTURE	WHAT IT ATTACHES OR DOES
Ovary	The almond-shaped female gonad beside the uterus. It makes oocytes and the hormones estrogen and progesterone
Ovarian ligament	Attaches the ovary to the uterus
Suspensory ligament	Attaches the ovary to the lateral pelvic wall. The ovarian artery and vein run in it
Broad ligament of the uterus	The fold of peritoneum that drapes over the uterus, uterine tubes, and ovaries and attaches them to the side walls of the pelvis
Round ligament of the uterus	A fibrous cord that runs from the uterus, through the inguinal canal, to the labia majora
Ovarian artery	Branches from the abdominal aorta and reaches the ovary through the suspensory ligament
Ovarian vein	Drains the ovary through the suspensory ligament

Ovary tissue, on the slide

The follicles are always in the cortex.

STRUCTURE	WHAT IT LOOKS LIKE	WHAT TO KNOW
Ovarian cortex	The outer region, crowded with follicles	Holds the developing oocytes
Ovarian medulla	The central region of looser connective tissue	Holds the blood vessels and nerves

STRUCTURE	WHAT IT LOOKS LIKE	WHAT TO KNOW
Ovarian follicles	An oocyte surrounded by one or more layers of follicle cells	Each holds an oocyte at a stage of development
Mature follicle	A large follicle with a big fluid-filled space and the oocyte off to one side	Ready to release its oocyte at ovulation
Corpus luteum	A large, solid, glandular body	Forms from the follicle after ovulation and makes progesterone and estrogen
Corpus albicans	A pale scar of dense connective tissue	What is left when the corpus luteum breaks down

The uterus

STRUCTURE	WHAT TO KNOW
Fundus	The rounded top of the uterus, above where the uterine tubes enter
Body	The main central part of the uterus
Cervix	The narrow lower part that projects into the vagina
Uterine cavity	The space inside the body of the uterus
Cervical canal	The channel through the cervix that joins the uterine cavity to the vagina
Perimetrium	The outer serous layer of the wall, part of the visceral peritoneum
Myometrium	The thick middle layer of smooth muscle that contracts during childbirth
Endometrium	The inner lining. Its functional layer sheds in menstruation and its basal layer rebuilds it each cycle

The vagina and the vulva

Vagina
The muscular canal from the cervix to the outside; the birth canal.

Vaginal orifice
The opening of the vagina, in the vestibule behind the urethral orifice.

Labia majora
The two larger outer folds of skin. They match the scrotum in the male.

Clitoris
The small mass of erectile tissue where the labia minora meet in front. It matches the penis in the male.

Urethral orifice
The opening of the urethra, between the clitoris and the vaginal orifice.

Fornix
The recess of the vagina that surrounds the cervix.

Mons pubis
The pad of fat over the pubic symphysis.

Labia minora
The two smaller, hairless folds inside the labia majora.

Vestibule
The area between the labia minora that holds the urethral and vaginal orifices.

Perineum
The diamond-shaped region between the thighs that holds the external genitalia and the anus.

The nervous tissue slides

On this sheet the cell you are pointing at is often named by the tissue it sits in.

SLIDE	WHAT IT SHOWS	WHAT TO IDENTIFY
Multipolar neuron	A large neuron with many processes, often from a spinal cord smear	Cell body, nucleus, nucleolus, dendrites, axon, axon hillock, Nissl bodies, neurofibrils
Spinal cord, cross section	A butterfly of gray matter in the center with white matter around it	Gray and white matter, ventral, dorsal, and lateral horns, central canal
Cerebral cortex	Gray matter on the outside of the cerebrum with white matter under it	Pyramidal cells, gray matter of the cortex, underlying white matter
Cerebellum	Folia with a branching white matter core	Purkinje cells, cerebellar cortex, arbor vitae
Nerve, cross section	Many axons bundled into fascicles and wrapped in connective tissue	Axons, Schwann cells, endoneurium, perineurium, epineurium, fascicles

The parts of a neuron

Cell body (soma)
The part of the neuron that holds the nucleus. It is the metabolic center of the cell.

Nucleolus
The dark dot inside the nucleus.

Axon
The single long process that carries the impulse away from the cell body.

Synaptic bulb (terminal end)
The swollen end of an axon branch that releases neurotransmitter at the synapse.

Neurofibrils
Fine threads of the cytoskeleton that run through the cell body and processes and give the neuron its shape.

Purkinje cells
Large multipolar neurons with a flask-shaped cell body and a huge branching tree of dendrites, found in the cerebellar cortex.

Nucleus
The large, pale round nucleus in the middle of the cell body.

Cell processes (dendrites)
The short branching processes that carry signals toward the cell body.

Axon hillock
The cone-shaped area where the axon leaves the cell body. It has no Nissl bodies, so it looks paler.

Nissl bodies
Dark-staining clumps of rough ER and ribosomes in the cell body, where proteins are made.

Pyramidal cells
Multipolar neurons with a triangle-shaped cell body, found in the cerebral cortex.

The parts of a nerve

Work from one axon outward. Nodes of Ranvier show on a lengthwise section, not on a cross section.

STRUCTURE	WHAT IT IS
Axon	A single nerve fiber
Schwann cells	Wrap the axons and form their myelin sheath in the PNS
Nodes of Ranvier	The gaps in the myelin sheath between two Schwann cells; the impulse jumps from node to node
Endoneurium	The innermost wrapping, around each individual axon
Perineurium	The middle wrapping, around each fascicle
Fascicle	A bundle of axons inside the nerve

STRUCTURE	WHAT IT IS
Epineurium	The outermost wrapping, around the whole nerve

The neuroglia

Four are in the CNS and two are in the PNS. Glial cells can still divide; neurons cannot.

GLIAL CELL	CNS OR PNS	WHERE YOU FIND IT	JOB
Astrocytes	CNS	Star-shaped cells between neurons and capillaries; the most numerous glia	Support neurons, help form the blood-brain barrier, and keep the chemical environment around neurons steady
Oligodendrocytes	CNS	Along axons, mostly in white matter	Form the myelin sheaths in the CNS; one cell wraps parts of several axons
Microglia	CNS	Small cells scattered through the brain and cord	Phagocytes that remove debris and microbes
Ependymal cells	CNS	Line the ventricles and the central canal and cover the choroid plexus	Help make and circulate cerebrospinal fluid
Schwann cells	PNS	Wrapped around peripheral axons	Form the myelin sheath in the PNS, one cell per segment of one axon, and help a cut axon regrow
Satellite cells	PNS	Around the neuron cell bodies in ganglia, such as the dorsal root ganglion	Surround, support, and protect those cell bodies

The same structure in the CNS and the PNS

STRUCTURE	IN THE CNS	IN THE PNS
A cluster of neuron cell bodies	Nucleus	Ganglion
A bundle of axons	Tract	Nerve
The cell that makes myelin	Oligodendrocyte	Schwann cell
The support cell around cell bodies	Astrocyte	Satellite cell

Surface landmarks of the cerebrum

LANDMARK	WHERE IT IS OR WHAT IT SEPARATES
Gyrus	A raised ridge of the cerebral cortex
Sulcus	A shallow groove between two gyri
Longitudinal fissure	The deep midline groove that separates the right and left cerebral hemispheres
Transverse fissure	The deep groove that separates the cerebrum from the cerebellum
Central sulcus	Separates the frontal lobe from the parietal lobe
Precentral gyrus	The gyrus just in front of the central sulcus; the primary motor cortex

LANDMARK	WHERE IT IS OR WHAT IT SEPARATES
Precentral sulcus	The groove at the front edge of the precentral gyrus
Postcentral gyrus	The gyrus just behind the central sulcus; the primary sensory cortex
Postcentral sulcus	The groove at the back edge of the postcentral gyrus
Lateral sulcus (Sylvian fissure)	Separates the temporal lobe below from the frontal and parietal lobes above; the insula lies deep inside it

The lobes of the cerebrum

Each hemisphere has all five lobes, so every lobe is paired. Four are named for the skull bone above them.

LOBE	WHERE IT IS	MAIN JOBS
Frontal lobe	In front of the central sulcus and above the lateral sulcus	Voluntary movement (precentral gyrus), speech production (Broca's area), planning, judgment, and personality
Parietal lobe	Behind the central sulcus and above the lateral sulcus, in front of the parieto-occipital sulcus	Touch, pain, temperature, and body position (postcentral gyrus), and spatial awareness
Temporal lobe	Below the lateral sulcus	Hearing, understanding language (Wernicke's area), and memory
Occipital lobe	At the back of the cerebrum, behind the parieto-occipital sulcus	Vision
Insula	Deep inside the lateral sulcus, covered by the frontal, parietal, and temporal lobes	Taste, sensation from the organs, and emotion

Midsagittal and inferior structures of the brain

Corpus callosum
The large C-shaped band of white matter that connects the right and left hemispheres.

Cingulate gyrus
The gyrus that arches over the corpus callosum. It is part of the limbic system and helps express emotion.

Septum pellucidum
The thin membrane under the corpus callosum that separates the two lateral ventricles.

Thalamus
The paired oval masses that make up most of the diencephalon; the main relay station for sensory signals on their way to the cortex.

Hypothalamus
The region below the thalamus that controls homeostasis: body temperature, hunger, thirst, the autonomic nervous system, and the pituitary gland.

Pineal gland
The small gland at the back of the diencephalon that secretes melatonin.

Sulcus of the corpus callosum
The groove between the corpus callosum and the cingulate gyrus.

Cingulate sulcus
The groove above the cingulate gyrus.

Fornix
The arch of white matter under the corpus callosum that runs from the hippocampus to the mammillary bodies.

Massa intermedia (interthalamic adhesion)
The small bridge that joins the right and left thalami across the third ventricle.

Infundibulum
The stalk that connects the hypothalamus to the pituitary gland.

Optic chiasm
The X where the two optic nerves meet, in front of the infundibulum. Fibers from the medial half of each retina cross here.

Optic nerve
Cranial nerve II, from the eye to the optic chiasm.

Mammillary bodies
Two small round bumps behind the infundibulum. They are part of the hypothalamus and work in memory.

Uncus
The hooked front end of the parahippocampal gyrus, on the medial temporal lobe. The primary olfactory cortex is here.

Olfactory bulb
The swelling under the frontal lobe, on the cribriform plate, where the olfactory nerve fibers end.

Optic tract
The continuation of the visual fibers from the optic chiasm back to the thalamus.

Crus cerebri
The front part of each cerebral peduncle, seen on the inferior surface of the midbrain. It carries descending motor tracts.

Parahippocampal gyrus
The gyrus on the medial side of the temporal lobe that covers the hippocampus; it works in memory.

Olfactory tract
The band that carries smell signals from the olfactory bulb back to the brain.

Functional areas of the cortex

Broca's and Wernicke's areas are usually in the left hemisphere.

AREA	WHERE IT IS	JOB
Primary motor cortex	Precentral gyrus of the frontal lobe	Starts voluntary movement of skeletal muscle on the opposite side of the body
Primary sensory cortex	Postcentral gyrus of the parietal lobe	Receives touch, pressure, pain, temperature, and body position from the opposite side of the body
Broca's area	Inferior frontal gyrus, usually the left	Controls the muscle movements of speech
Wernicke's area	Back of the superior temporal gyrus, where the temporal lobe meets the parietal lobe, usually the left	Understanding spoken and written language
Thalamus	Diencephalon, deep to the cerebrum (not part of the cortex)	Relays sensory signals to the cortex

The cerebellum

Cerebellum
The second largest part of the brain, behind the brainstem and below the occipital lobes. It coordinates movement, posture, and balance.

Vermis
The narrow midline strip that joins the two cerebellar hemispheres.

Cerebellar cortex
The outer layer of gray matter. The Purkinje cells are here.

Transverse fissure
The deep groove between the cerebrum and the cerebellum.

Cerebellar hemispheres
The right and left halves of the cerebellum.

Folia
The fine, leaf-like folds on the surface of the cerebellum.

Arbor vitae
The branching, tree-shaped white matter inside the cerebellum, best seen on a midsagittal cut.

Arteries of the brain

The middle cerebral artery is not part of the ring.

ARTERY	COMES FROM	WHAT IT SUPPLIES
Internal carotid artery	Common carotid artery in the neck	Most of the cerebrum. It ends by dividing into the anterior and middle cerebral arteries
Anterior cerebral artery	Internal carotid artery	Medial surface of the frontal and parietal lobes, including the motor and sensory cortex for the lower limb
Anterior communicating artery	Joins the right and left anterior cerebral arteries	Closes the front of the ring
Middle cerebral artery	Internal carotid artery, its largest branch; it runs in the lateral sulcus	Lateral surface of the cerebrum, including the motor and sensory cortex for the face and upper limb
Posterior communicating artery	Joins the internal carotid artery to the posterior cerebral artery	Links the front and back blood supplies
Posterior cerebral artery	End branches of the basilar artery	Occipital lobe and the inferior surface of the temporal lobe
Basilar artery	Formed where the two vertebral arteries join; runs up the front of the pons	Brainstem and cerebellum through its branches
Pontine arteries	Small branches of the basilar artery	Pons
Vertebral arteries	Subclavian arteries; they enter the skull through the foramen magnum	Join to form the basilar artery
Superior cerebellar artery	Top of the basilar artery, just below the posterior cerebral artery	Upper surface of the cerebellum
Anterior inferior cerebellar artery	Lower part of the basilar artery	Front of the lower surface of the cerebellum
Posterior inferior cerebellar artery	Vertebral artery	Lower surface of the cerebellum and the side of the medulla
Anterior spinal artery	Branches of both vertebral arteries that join in the midline	Front of the spinal cord

The brainstem regions

Name the region before you name the part.

REGION	WHERE IT IS	WHAT TO IDENTIFY ON IT	CRANIAL NERVES
Midbrain	The top of the brainstem, just below the diencephalon	Cerebral peduncles, with the crus cerebri on the front; corpora quadrigemina (superior and inferior colliculi) on the back; the cerebral aqueduct running through it; the superior cerebellar peduncle	III and IV
Pons	The rounded bulge in the middle of the brainstem, in front of the cerebellum	Basilar pons, the bulging front of the pons; the fourth ventricle behind it; the middle cerebellar peduncle	V, VI, VII, and VIII. V attaches to the side of the pons; VI, VII, and VIII leave at the pontomedullary junction
Pontomedullary junction	The groove where the pons meets the medulla	The groove itself, and the nerves that leave from it	VI, VII, and VIII
Medulla oblongata	The bottom of the brainstem; it becomes the spinal cord at the foramen magnum	Pyramids, decussation of the pyramids, the medullary olive beside each pyramid, the inferior cerebellar peduncle, and the lower part of the fourth ventricle	IX, X, XI, and XII (most of the fibers of XI come from the upper cervical spinal cord)

Structures of the brainstem

Cerebral peduncles
The two thick stalks on the front of the midbrain that carry motor tracts down from the cerebrum.

Cerebral aqueduct
The narrow channel through the midbrain that carries CSF from the third ventricle to the fourth ventricle.

Superior colliculi
The upper pair; reflex centers that turn the eyes and head toward something you see.

Basilar pons
The bulging front part of the pons. It holds the pontine nuclei, which relay signals from the cerebral cortex to the cerebellum.

Pyramids
Two ridges on the front of the medulla, formed by the corticospinal (voluntary motor) tracts.

Medullary olive
The oval bulge on each side of the medulla, just lateral to the pyramid. It holds a nucleus that relays signals to the cerebellum.

Crus cerebri
The front part of each cerebral peduncle, made of descending motor fibers.

Corpora quadrigemina
The four bumps on the back of the midbrain: two superior colliculi and two inferior colliculi.

Inferior colliculi
The lower pair; reflex centers that turn the head toward a sound.

Fourth ventricle
The CSF space behind the pons and upper medulla, in front of the cerebellum.

Decussation of the pyramids
The spot at the lower end of the pyramids where most corticospinal fibers cross to the opposite side. This is why each side of the brain controls the opposite side of the body.

The cerebellar peduncles

Superior to the midbrain, middle to the pons, inferior to the medulla.

PEDUNCLE	CONNECTS THE CEREBELLUM TO	MAIN TRAFFIC
Superior cerebellar peduncle	Midbrain	Mostly output from the cerebellum
Middle cerebellar peduncle	Pons	Input from the pontine nuclei, carrying signals from the cerebral cortex
Inferior cerebellar peduncle	Medulla oblongata	Input from the spinal cord and the balance (vestibular) system

The meninges

From the skull inward: dura mater, arachnoid mater, pia mater.

LAYER	POSITION	WHAT TO KNOW
Dura mater	The outer layer	Tough and thick. In the skull it has two layers that separate to form the dural venous sinuses
Periosteal layer of the dura	The outer layer of the cranial dura, attached to the inside of the skull	Acts as the periosteum of the skull bones. It is not present around the spinal cord
Meningeal layer of the dura	The inner layer of the cranial dura	Folds inward to form the falx cerebri, falx cerebelli, and tentorium cerebelli; continues down as the spinal dura
Arachnoid mater	The middle layer	A thin, web-like membrane. Its arachnoid granulations push into the dural sinuses

LAYER	POSITION	WHAT TO KNOW
Pia mater	The inner layer	A delicate layer full of blood vessels that clings to the surface of the brain and cord and follows every gyrus and sulcus

The dural extensions

EXTENSION	WHERE IT IS	WHAT IT SEPARATES
Falx cerebri	In the longitudinal fissure	The right and left cerebral hemispheres
Falx cerebelli	In the midline below the tentorium cerebelli	The right and left cerebellar hemispheres
Tentorium cerebelli	In the transverse fissure, like a tent over the cerebellum	The cerebrum above from the cerebellum below

The meningeal spaces

SPACE	BETWEEN	WHAT IS IN IT
Epidural space	The dura mater and the bone	In the vertebral canal: fat and veins. Inside the skull there is no epidural space, because the dura is attached to the bone
Subdural space	The dura mater and the arachnoid mater	Only a potential space, with a thin film of fluid
Subarachnoid space	The arachnoid mater and the pia mater	Cerebrospinal fluid and the large blood vessels of the brain
Arachnoid granulations	Projections of the arachnoid mater into the superior sagittal sinus	Where CSF is reabsorbed into the venous blood

The ventricles

STRUCTURE	WHERE IT IS	WHAT TO KNOW
Lateral ventricle	One in each cerebral hemisphere, under the corpus callosum	Connects to the third ventricle through the interventricular foramen
Septum pellucidum	Under the corpus callosum, in the midline	The thin wall that separates the two lateral ventricles
Interventricular foramen of Monro	The small opening at the front of the third ventricle	Joins each lateral ventricle to the third ventricle
Third ventricle	The narrow midline space in the diencephalon, between the two thalami	Connects to the fourth ventricle through the cerebral aqueduct
Cerebral aqueduct	The narrow channel through the midbrain	Joins the third ventricle to the fourth ventricle
Fourth ventricle	Between the pons and upper medulla in front and the cerebellum behind	Opens into the subarachnoid space through the median aperture and two lateral apertures; continuous below with the central canal
Central canal	The tiny channel down the center of the spinal cord	Continuous above with the fourth ventricle; holds a small amount of CSF
Choroid plexus	Tufts of capillaries covered by ependymal cells, in the lateral, third, and fourth ventricles	Makes the CSF

External anatomy of the spinal cord

The cord ends at about the L1 to L2 level in an adult, well above the end of the vertebral column.

STRUCTURE	WHAT TO KNOW
Cervical enlargement	The wider part of the cord in the neck (about C4 to T1) where the nerves to the upper limbs arise
Lumbar enlargement	The wider part of the lower cord where the nerves to the lower limbs arise
Conus medullaris	The cone-shaped end of the cord, at about the disc between L1 and L2
Cauda equina	The bundle of nerve roots that hangs below the conus medullaris in the vertebral canal, like a horse's tail. A lumbar puncture needle goes in here, between L3 and L4 or L4 and L5, below the end of the cord
Filum terminale	A thread of pia mater that runs from the conus medullaris down to the coccyx and anchors the cord
Denticulate ligaments	Tooth-shaped extensions of pia mater along each side of the cord that anchor it to the dura
Anterior median fissure	The deep, wide groove down the front of the cord
Posterior median sulcus	The shallow, narrow groove down the back of the cord

The spinal cord in cross section

In the cord the gray matter is inside and the white matter is outside, the reverse of the cerebrum.

STRUCTURE	WHAT IS IN IT
Gray matter	The butterfly-shaped core of neuron cell bodies, dendrites, and unmyelinated axons
Ventral (anterior) horn	Cell bodies of somatic motor neurons, whose axons go to skeletal muscle
Lateral horn	Cell bodies of autonomic (sympathetic) motor neurons; present only in segments T1 to L2
Dorsal (posterior) horn	Interneurons and the incoming axons of sensory neurons
Gray commissure	The bar of gray matter that joins the two sides, around the central canal
Central canal	The tiny CSF channel in the center of the gray commissure, continuous with the fourth ventricle
Anterior white commissure	The band of white matter in front of the gray commissure where axons cross from one side to the other
White matter	Myelinated axons running up and down the cord in tracts
Anterior column	White matter between the anterior median fissure and the ventral horn
Lateral column	White matter on the side of the cord, between the dorsal and ventral horns
Posterior column	White matter between the dorsal horn and the posterior median sulcus; carries ascending sensory tracts

The spinal nerve and its parts

Everything before the spinal nerve is a root; everything after it is a ramus. Only the dorsal root has a ganglion.

STRUCTURE	WHERE IT IS	WHAT IT CARRIES
Rootlets	The fine bundles that attach to the cord in a row and join to form a root	Sensory axons (dorsal rootlets) or motor axons (ventral rootlets)
Dorsal (posterior) root	Joins the back of the cord	Sensory axons coming into the cord
Dorsal root ganglion	The swelling on the dorsal root, near the intervertebral foramen	Sensory cell bodies
Ventral (anterior) root	Leaves the front of the cord	Motor axons going out to muscles and glands
Spinal nerve	Where the dorsal and ventral roots join	Mixed: sensory and motor axons

STRUCTURE	WHERE IT IS	WHAT IT CARRIES
Intervertebral foramen	The opening between two vertebrae	The spinal nerve leaves the vertebral canal through it
Dorsal (posterior) ramus	The branch that goes to the back	Mixed; serves the deep muscles and skin of the back
Ventral (anterior) ramus	The larger branch that goes to the front and sides	Mixed; serves the limbs and the front and sides of the trunk, and forms the plexuses
White ramus communicans	Connects the ventral ramus to a sympathetic chain ganglion; only at T1 to L2	Myelinated sympathetic axons going from the cord to the chain ganglion (preganglionic)
Gray ramus communicans	Connects a sympathetic chain ganglion back to the spinal nerve; at every spinal level	Unmyelinated sympathetic axons going out from the chain ganglion to the body (postganglionic)
Sympathetic chain ganglion	Beside the vertebral body	Cell bodies of postganglionic sympathetic neurons
Sympathetic trunk	The chain of sympathetic ganglia linked together along each side of the vertebral column	Sympathetic axons running up and down between the ganglia

The plexuses

Only the ventral rami form plexuses. The ventral rami of T2 to T12 do not form a plexus; they run as the intercostal nerves. The phrenic nerve is the nerve on the Module 5 list; the other nerves here are from your earlier lab lists.

PLEXUS	SPINAL NERVES	REGION IT SERVES	MAIN NERVES
Cervical plexus	C1 to C4 (C5 adds fibers to the phrenic nerve)	Skin and muscles of the neck and upper shoulder, and the diaphragm	Phrenic nerve (C3, C4, C5), the motor nerve to the diaphragm
Brachial plexus	C5 to T1	The shoulder and the whole upper limb	Musculocutaneous, median, ulnar, radial, and axillary nerves
Lumbar plexus	L1 to L4	The lower abdominal wall, the external genitals, and the front and inner thigh	Femoral and obturator nerves
Sacral plexus	L4 to S4	The buttock, the perineum, the back of the thigh, and most of the leg and foot	Sciatic nerve, which splits into the tibial and common fibular nerves

The cranial nerves

Order of the names: On Old Olympus' Towering Tops A Finn And German Viewed Some Hops. Sensory, motor, or both: Some Say Marry Money But My Brother Says Big Brains Matter Most.

NUMBER	NAME	SENSORY, MOTOR, OR BOTH	WHERE IT LEAVES THE SKULL	MAIN JOB
CN I	Olfactory	Sensory	Olfactory foramina in the cribriform plate of the ethmoid	Smell
CN II	Optic	Sensory	Optic canal	Vision
CN III	Oculomotor	Motor	Superior orbital fissure	Moves the eye with four of the extrinsic eye muscles, raises the upper eyelid, and constricts the pupil
CN IV	Trochlear	Motor	Superior orbital fissure	Moves the eye with the superior oblique

NUMBER	NAME	SENSORY, MOTOR, OR BOTH	WHERE IT LEAVES THE SKULL	MAIN JOB
CN V	Trigeminal	Both	Superior orbital fissure (V1), foramen rotundum (V2), foramen ovale (V3)	Sensation from the face, scalp, and teeth, and movement of the muscles of chewing
CN VI	Abducens	Motor	Superior orbital fissure	Moves the eye laterally with the lateral rectus
CN VII	Facial	Both	Internal acoustic meatus, then out through the stylomastoid foramen	Facial expression, taste from the front two thirds of the tongue, and tears and saliva
CN VIII	Vestibulocochlear	Sensory	Internal acoustic meatus (it stays in the temporal bone to reach the inner ear)	Hearing and balance
CN IX	Glossopharyngeal	Both	Jugular foramen	Taste and sensation from the back third of the tongue, swallowing, and saliva from the parotid gland
CN X	Vagus	Both	Jugular foramen	Parasympathetic control of the heart, lungs, and digestive organs, plus swallowing and the voice
CN XI	Accessory	Motor	Jugular foramen	Moves the sternocleidomastoid and trapezius
CN XII	Hypoglossal	Motor	Hypoglossal canal	Moves the tongue

Structures associated with the cranial nerves

STRUCTURE	WHAT IT IS	NERVE
Olfactory foramina in the cribriform plate	The tiny holes in the cribriform plate of the ethmoid	CN I fibers pass up through them to the olfactory bulb
Olfactory bulb	The swelling on the cribriform plate where the olfactory nerve fibers end	CN I
Olfactory tract	Carries smell signals from the olfactory bulb back to the brain	CN I pathway
Optic canal	The opening in the sphenoid at the back of the orbit	CN II
Optic nerve	The nerve from the back of the eye to the optic chiasm	CN II
Optic chiasm	Where the two optic nerves meet and fibers from the medial half of each retina cross	CN II pathway
Optic tract	Carries visual fibers from the optic chiasm back to the thalamus	CN II pathway
Superior orbital fissure	The slit between the greater and lesser wings of the sphenoid, at the back of the orbit	CN III, IV, V1, and VI
Internal acoustic meatus	The opening on the back of the petrous part of the temporal bone	CN VII and VIII
Pontomedullary junction	The groove between the pons and the medulla	CN VI, VII, and VIII leave the brainstem here

PART C

Questions you could be asked on the lab exam.

At each station a structure will be pinned or pointed to. These are the kinds of questions that go with it. Practice answering them out loud with a partner, with the structure in front of you.

The Urinary System

1. Identify the structure at the pointer on the kidney model.
2. Which kidney sits lower, and why?
3. Name the region at the pointer: renal cortex, renal medulla, renal column, or renal pyramid.
4. Is the cup at the pointer a minor calyx or a major calyx, and what drains into it?
5. Name the blood vessel at the pointer. Does it carry blood toward the glomerulus or away from it?
6. Trace a drop of blood from the renal artery to the renal vein, naming every vessel.
7. Name the part of the renal corpuscle at the pointer: glomerulus, parietal layer, visceral layer (podocytes), or capsular space.
8. Name the segment of the renal tubule at the pointer.
9. Trace the filtrate from the glomerular capsule to the outside of the body.
10. Name the two cell types of the juxtaglomerular apparatus and what each one does.
11. Name the smooth triangle on the floor of the bladder and the three openings at its corners.
12. Which urethral sphincter is under voluntary control, and what kind of muscle is it?

The Male Reproductive System

1. Identify the structure at the pointer.
2. Which muscle wrinkles the scrotal skin, and which raises the testis? Which one is smooth muscle and which is skeletal?
3. Name the covering of the testis at the pointer: tunica vaginalis or tunica albuginea.
4. On the testis slide, identify the cell at the pointer: sustentacular (Sertoli) cell, interstitial (Leydig) cell, or developing sperm cell.
5. Which cells make testosterone, and where are they on the slide?
6. Name three structures that travel in the spermatic cord.
7. Which structure makes sperm, and which stores them while they gain the ability to swim?
8. Trace the path of sperm from the seminiferous tubules to the outside of the body.
9. Name the part of the urethra at the pointer: prostatic, membranous, or spongy.
10. Name the accessory gland at the pointer and what it adds to semen.
11. Name the column of erectile tissue at the pointer. Which one surrounds the urethra?
12. The medial umbilical ligament is the remnant of which fetal structure?

The Female Reproductive System

1. Identify the structure at the pointer.
2. Name the ligament at the pointer and what it attaches.
3. On the ovary slide, is the region at the pointer the cortex or the medulla?
4. Identify the structure on the ovary slide: ovarian follicle, mature follicle, corpus luteum, or corpus albicans.
5. What does the corpus luteum become when it breaks down?
6. Name the region of the uterine tube at the pointer. In which region does fertilization usually happen?
7. Trace the path of an oocyte from the ovary to the uterine cavity.
8. Name the region of the uterus at the pointer: fundus, body, or cervix.
9. Name the layer of the uterine wall at the pointer. Which layer sheds in menstruation, and which layer rebuilds it?
10. Where does the round ligament of the uterus run, and where does it end?
11. Name the structure of the vulva at the pointer.
12. Is the artery at the pointer the internal or the external iliac, and what does it supply?

Nervous Tissue, Histology

1. Identify the tissue on this slide: multipolar neuron, spinal cord, cerebral cortex, cerebellum, or nerve.
2. Name the part of the neuron at the pointer.
3. What are the dark clumps in the cell body at the pointer called, and what do they do?
4. Name the cone-shaped area where the axon leaves the cell body.
5. Identify the large neuron at the pointer on the cerebellum slide.
6. Identify the triangle-shaped neuron at the pointer on the cerebral cortex slide.
7. On the spinal cord cross section, is the region at the pointer gray matter or white matter?
8. Name the connective tissue layer at the pointer on the nerve: endoneurium, perineurium, or epineurium. What does it surround?
9. Name the glial cell at the pointer. Is it found in the CNS or the PNS, and what does it do?
10. Which cell builds the myelin in the PNS, and which one builds it in the CNS?
11. What is a cluster of cell bodies called in the CNS and in the PNS? What is a bundle of axons called in each?

The Brain

1. Identify the gyrus, sulcus, or fissure at the pointer.
2. Which gyrus lies immediately anterior to the central sulcus, and what does it do?
3. Which gyrus lies immediately posterior to the central sulcus, and what does it do?
4. Name the lobe at the pointer and the sulcus or fissure that marks its border.
5. Name the lobe hidden deep in the lateral sulcus.
6. On the midsagittal section, identify the structure at the pointer.
7. What does the corpus callosum connect, and what thin membrane hangs beneath it?
8. Name the stalk that connects the hypothalamus to the pituitary gland.
9. On the inferior surface, identify the structure at the pointer: olfactory bulb, optic chiasm, mammillary bodies, crus cerebri, or uncus.
10. Name the functional area at the pointer and the gyrus it sits on. In which hemisphere are Broca's and Wernicke's areas usually found?
11. Identify the artery at the pointer in the cerebral arterial circle.
12. Trace the route from a vertebral artery into the ring and around to an anterior cerebral artery, naming every vessel on the way.

The Brainstem and Cerebellum

1. Name the region of the brainstem at the pointer: midbrain, pons, or medulla oblongata.
2. Identify the structure at the pointer on the brainstem.
3. Which cranial nerves arise from the midbrain, the pons, and the medulla?
4. Which cranial nerves leave the brainstem at the pontomedullary junction?
5. What crosses at the decussation of the pyramids, and what does that crossing explain?
6. Name the oval bulge at the pointer, just lateral to the pyramid.
7. Which colliculi are reflex centers for sight, and which are reflex centers for sound?
8. Name the cerebellar peduncle at the pointer and the region of the brainstem it connects to.
9. Name the branching white matter at the pointer inside the cerebellum.
10. Identify the part of the cerebellum at the pointer: hemisphere, vermis, folia, or cerebellar cortex.
11. Which region of the brainstem is continuous with the spinal cord, and at what landmark?

Meninges, Ventricles, and CSF

1. Name the meningeal layer at the pointer, and name all three layers in order from the skull inward.
2. Name the two layers of the cranial dura mater.
3. Name the dural extension at the pointer, where it is found, and the structures it separates.
4. Identify the dural venous sinus at the pointer.
5. Where does the sigmoid sinus leave the skull, and what does the blood become once it does?
6. Name the space at the pointer. Which space holds CSF, and which is only a potential space?
7. Identify the ventricle at the pointer.
8. Name the opening that joins a lateral ventricle to the third ventricle, and the channel that joins the third ventricle to the fourth.
9. Name the thin membrane that separates the two lateral ventricles.
10. What structure makes CSF, and which cells cover it?
11. Trace cerebrospinal fluid from the lateral ventricles to its reabsorption, naming every structure it passes, including the median and lateral apertures and the central canal.
12. Where is CSF reabsorbed into the blood?

The Spinal Cord

1. Name the meningeal layer or space at the pointer.
2. What fills the epidural space in the vertebral canal, and why is there no epidural space inside the cranium?
3. Name the structures that anchor the sides of the cord to the dura.
4. Identify the structure at the pointer: cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, or filum terminale.
5. Why is the cord wider at the cervical and lumbar enlargements?
6. At what vertebral level does the cord end, and what fills the canal below that point?
7. Where in the vertebral column is a lumbar puncture performed, and why is that level safe?
8. Is the groove at the pointer the anterior median fissure or the posterior median sulcus? Which side of the cord are you looking at?
9. Name the horn at the pointer and what its cell bodies do.
10. Which segments of the cord have a lateral horn, and what sits in it?
11. Name the white column at the pointer: anterior, lateral, or posterior.
12. Name the channel in the center of the gray commissure and what fills it.

Peripheral Nerves and Plexuses

1. Identify the structure at the pointer: rootlets, dorsal root, ventral root, dorsal root ganglion, or spinal nerve.
2. What does the structure at the pointer carry: sensory axons, motor axons, mixed, or sensory cell bodies?
3. Which root carries a ganglion, and whose cell bodies are in it?
4. Through what opening does the spinal nerve leave the vertebral canal?
5. Name the ramus at the pointer. Which ramus forms the plexuses, and which one serves the deep muscles and skin of the back?
6. Is the ramus communicans at the pointer white or gray, and what does it carry?
7. Identify the sympathetic chain ganglion and the sympathetic trunk.
8. Name the plexus at the pointer and the spinal nerves that form it.
9. Name the plexus the phrenic nerve comes from and the muscle it supplies.
10. A patient loses sensation and strength across one whole upper limb. Which plexus is damaged?
11. Which plexus serves the buttock and most of the lower limb?

The Cranial Nerves

1. Identify the cranial nerve at the pointer by number and name.
2. Is the nerve at the pointer sensory, motor, or both?
3. What is the main job of the nerve at the pointer?
4. Name the opening in the skull that the nerve at the pointer passes through.
5. Which nerves pass through the superior orbital fissure? Which pass through the internal acoustic meatus?
6. Name the structure at the pointer: olfactory bulb, olfactory tract, optic nerve, optic chiasm, or optic tract.
7. Through which openings do the olfactory nerve fibers pass to reach the olfactory bulb?
8. Identify the extrinsic eye muscle at the pointer and the cranial nerve that supplies it.
9. Which muscle moves the eye laterally, and which nerve supplies it?
10. Which three cranial nerves are purely sensory?
11. Which four cranial nerves are mixed?
12. Which cranial nerve moves the sternocleidomastoid and trapezius, and which one moves the tongue?